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COMMENTS AND CORRECTIONS

Corrections to "Enhancing Knapsack-Based Financial Portfolio Optimization Using Quantum Approximate Optimization Algorithm"

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In the above article [1], one reference [2] was missing. Insert at the end of Section I, immediately before the contribution lists: "Our work presents a QAOA-based approach for solving the portfolio optimization problem by reformulating it as a knapsack problem. While certain quantum circuit components, such as the feasibility oracle and mixer structure, are consistent with prior constructions [2], the novelty of our study lies in extending this framework to the domain of financial portfolio selection. We map expected returns from Markowitz analysis to the knapsack formulation, incorporate domain specific investment constraints, and evaluate the method using both simulated and real quantum devices. This adaptation illustrates how QAOA-based methods can be applied in practice to complex financial decision problems.

One sentence is appended at the end of Code Availability section: "The authors acknowledge the Bachelor thesis by Jan Rasmus Holst, [2], which served as a valuable reference for designing the feasibility oracle and mixer components within the QAOA framework."

The authors requested this correction to complement the missing references as well as explaining why the above article [1] has a similar part in mathematical derivation with reference [2].

REFERENCES

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